Assignment 2

BRAC University (Department of Computer Science and Engineering) CSE

CSE340 Summer26

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1. For opcode , func3 and func7 use don't care(××)

Instruction	Machine Code of the corresponding instruction.
Label0: SUB X10, X9, X23	
Label1: BNE X24, X25, Label3	
Label2: ADD X5, X5, X6	
SUB X5, X22, X5	
Label3: SLLI X22, X22, 3	
ADD X22, X22, X23	
Label4: LD X6, 32(X22)	
BEQ X4, X8, _____	0xFE825A80
Label5: SW X7, 8(X21)	

Find the correct label to be placed in the blank. Answer this in your script. You must show the necessary calculations in your script.

2. Construct the equivalent RISC-V code of the following C code.

```
if ( A[3] != A[6]){
    if (A[3] == 0) {
        A[3] = A[3] + 2;
    }else{
        A[6] = A[6] / 16;
    }
}else{
```

A[6] = A[6] * 8
}

Base addresses of array A and B are in register X_{20} and X_{21} .

3.

Translate the following codes written in high level language into an instruction sequence written in RISC-V Assembly. Assume that variables i, j, and n are stored in registers X21, X22, and X23. A and B are byte arrays. A[0] and B[0] are stored in X5 and X6 respectively.

```
if (A[i] > 3) {
     if (B[j] == 4) {
          B[j] += A[i]
}
else {
     B[j] -= 2
}
}
```

```
i = 0;
n = 5;
while (i < n) {
     i += 1;
}
```